- H1047R and C420 seem to attack the brain differently
- H1047R appears to hijack the circuits electrically by expressing synapse building genes and plug itself into the brain's network to hijack its signals
- This could force neurons to extensively fire (triggering seizures), causes metabolic collapse, shutting down mitochondrial respiration and ATP energy grid

  Data supporting this (malignant - control vs H1047R, GSEA)

  - Cell-Cell Adhesion via Plasma-Membrane molecules +2.31 NES
  - Modulation of Chemical Synaptic Transmission +2.29 NES
  - Synapse organization +2.19 NES

  Data supporting this (neurons - H1047R vs C420R, GSEA)

  - Aerobic Respiration -3.01 NES
  - Oxidative Phosphorylation - 2.97 NES
  - Mitochondrial ATP Synthesis -2.82 NES
  - Mitochondrial Translation -2.61 NES

  Data supporting this (malignant - H1047R classification)

  - Slc1a2 (Mean Abs SHAP = 0.0122)
  - Egr1 (Mean Abs SHAP = 0.0104)

  Data supporting this (neurons - H1047R classification)

  - Egr1 (Mean Abs SHAP = 0.0079)
  - Fos (Mean Abs SHAP = 0.0048)

- C420R seemingly doesn't build synapses but hijacks actomyosin contractile pathways (same molecular motors used in muscle cells) to compress the brain tissue

  Data supporting this: (malignant - control vs C420R):

  - Striated muscle contraction +2.81 NES
  - Skeletal muscle contraction +2.59 NES
  - Muscle contraction +2.43 NES
  - Myofibrial assembly +2.38 NES

  Data supporting this: (neurons - C420R vs R88Q):

  - Muscle contraction +2.82 NES
  - Myofibril assembly +2.74 NES
  - Striated Muscle Contraction +2.71 NES

  Data supporting this: (malignant - C420R classification)

- Tsmb4x (Mean Abs SHAP = 0.0052)
- Malat1 (Mean Abs SHAP = 0.0080)

Data supporting this (neurons - C420R classification)

- Csrp1 (Mean Abs SHAP = 0.0047)
- Thrsp (Mean Abs SHAP = 0.0040)

Note: The two methods (GSAP vs ML/SHAP) are generally independent, furthermore GSAP is linear and ML/SHAP (random forest classifier) non-linear. They generally support one another's conclusions. There is however a dependent part which is the data given to these models.